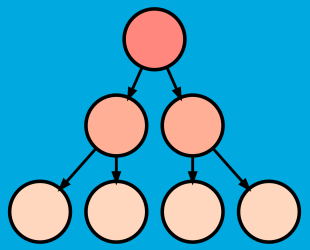
# Multilevel Modeling in Clinical Research



## Concepts and Applications

Waylon Howard

Webinar, March 18, 2026

# About me

Waylon Howard (waylon.howard@seattlechildrens.org)

- Principal Biostatistician and Quantitative Methodologist at: *Biostatistics, Epidemiology and Analytics in Research (BEAR)* (SCRI)
  - Research methods, statistics, and measurement
  - Grant writing, analysis planning, and manuscripts
  - Methodological innovation and applied solutions

**I help researchers answer the right questions!**

# Preliminaries

Slides and material are available at

https://github.com/wwwaylon/chbd-lgm-2025

- The talk consists of a combination of slides and hands-on exercises
- 💬 Feel free to ask questions anytime!
- We will primarily rely on `R`, `RStudio` & `GitHub`.

**For my research, I typically use Mplus and R (`lmer`).**

# Software for Multilevel Modeling

| Software | MLM Module |
|---|---|
| R | `lme4`, `nlme`, `xxM`, `DESeq2` |
| SAS | `PROC MIXED` |
| SPSS | `MIXED` |
| Stata | `xtmixed` / `mixed` |
| LISREL | `MULTILEV` |
| HLM | Purpose-built, widely cited |
| MLwiN | Strong for complex structures |
| EQS | Built-in |
| S-Plus | Built-in |
| SYSTAT | Built-in |
| Mplus | SEM + MLM, highly flexible |

## My Recommendations

### For research:

> Mplus for flexibility with SEM and complex models
>
> R (`lme4`) for reproducibility and custom extensions

No single tool is best for every situation - choose based on model complexity, team expertise, and reproducibility needs. Also consider factors like collaboration, and cost.

# Many names - same general idea

## Model frameworks

> Multilevel models, Hierarchical linear models (HLM), Mixed-effects models, Linear mixed models (LMM), Growth curve models, Random coefficient models

## Model components

> Variance components, Random effects / coefficients, Random intercepts and slopes, Varying coefficients, Intercepts- or slopes-as-outcomes, *Cross-level interactions*, *Contextual effects*

👉 These are all variations of mixed models.

# Plan Today

| Time | Segment | Key Concepts | Slides |
|---|---|---|---|
| 0–10 | Why is this important? | Nested data in the real world | 1–6 |
| 10–25 | Estimation in Statistical Models | OLS vs ML estimation | 7–15 |
| 25–40 | How not to handle nested data | Traditional methods & limitations | 16–25 |
| 40–55 | Empty means models & ICC | Variance decomposition<br>Empty model & ICC | 26–35 |
| 55–85 | Conditional models & Effect Size | Model building & effect size | 36–52 |
| 85–90 | Next Steps & Extensions | Extensions & resources | 53–55 |

# Why multilevel models?

- Data are clustered ➡️ observations are related
- Standard analyses assume independence

Example: hospital re-admissions

> *Patients in the same hospital share unmeasured factors.*
> *Repeated measurements on the same patient are not independent.*

**Violations of this assumption bias standard errors.**

# Kinds of Data Structure

- **Hierarchical (Nested)** repeated tests - students - schools - districts
- **Crossed (Non-nested)** observations within participants *and* tasks
- **Balanced vs. Unbalanced** Experiments vs. Real-world
- **Other Structures** Temporal (time-ordered), Spatial (location-based)

Once you learn about hierarchical data, you start seeing examples everywhere!

*Image adapted from Preacher (2010) lecture - thanks Kris.*

# 🎯 Estimation in Statistical Models

## How do statistical models estimate parameters?

| Model | Outcome | Effects | Sampling |
|---|---|---|---|
| Linear Model | Normal | Fixed | 1 level |
| Generalized Linear Model | Any | Fixed | 1 level |
| Linear *Mixed* Model | Normal | Fixed + Random | 2+ levels |
| Generalized Linear *Mixed* Model | Any | Fixed + Random | 2+ levels |

### 🔑 Key question:

> How do we estimate the parameters of these models?

We start with Ordinary Least Squares (OLS)

# 🎯 Estimation in Statistical Models

## Ordinary Least Squares (OLS)

$$\text{Linear regression: } \mathrm{y}_i = \beta_0 + \beta_1 \mathrm{x}_i + \mathrm{e}_i$$

$$\text{Residual: } \mathrm{e}_i = \mathrm{y}_i - \hat{\mathrm{y}}_i$$

$$\text{OLS chooses coefficients that minimize: } \sum_{i=1}^{n} (\mathrm{y}_i - \hat{\mathrm{y}}_i)^2$$

## 💡 Core idea

> Find the line that minimizes the total squared prediction error.

# 🎯 Estimation in Statistical Models

This scatterplot shows the relationship between a predictor X and an outcome Y. Each orange point represents an observed score.

**OLS will search for the line that best predicts these values.**

# 🎯 Estimation in Statistical Models

**Features of the best-fitting line:** Intercept $\beta_0$, Slope $\beta_1$, Residual variance

These parameters define the model that best predicts the data.

# 🎯 Estimation in Statistical Models

A residual is the vertical distance to the regression line: $\mathrm{e}_i = \mathrm{y}_i - \hat{\mathrm{y}}_i$

But OLS does not minimize the raw residuals.

# 🎯 Estimation in Statistical Models

## Why Square Residuals?

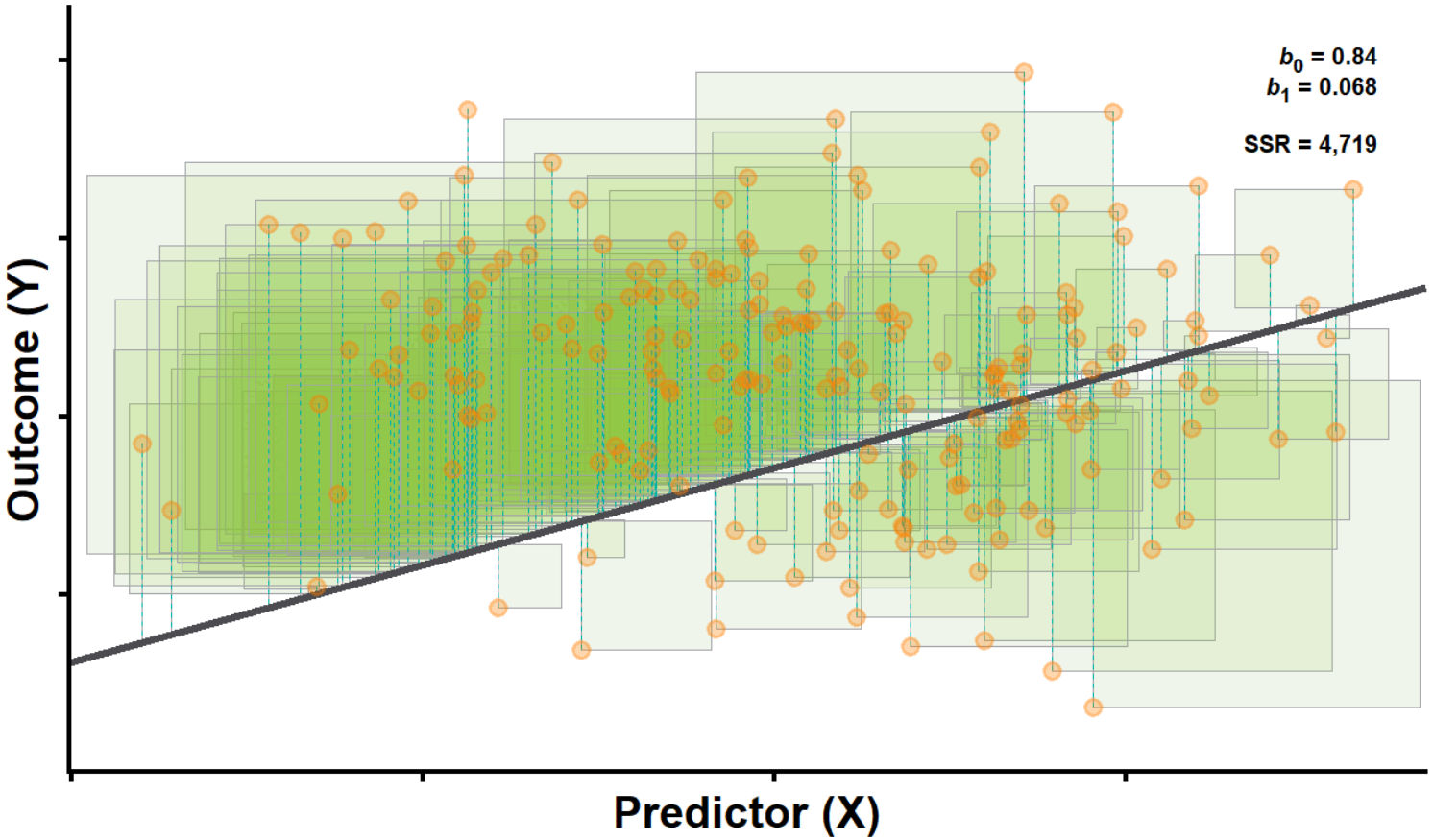
| Strategy | Problem |
|---|---|
| Minimize **sum** of residuals | Positive & negative cancel |
| Minimize **sum of absolute** residuals | Difficult to solve |
| Minimize **sum of squared** residuals | Works 🏆 ✨ |

Key benefits:

- ✅ makes all values positive
- ✅ penalizes large errors
- ✅ produces a **closed-form solution**

# 🎯 Estimation in Statistical Models

OLS chooses the line that minimizes: $\sum_{i=1}^{n}(\mathrm{y}_i - \hat{\mathrm{y}}_i)^2$




Geometrically:

> OLS minimizes the total area of the residual squares.

# 🎯 Estimation in Statistical Models

The squared-error function forms a **smooth quadratic surface.**

This means: ✅ there is **one optimal solution** ✅ we can solve it analytically

**OLS therefore has a closed-form estimator.**

# 🎯 Estimation in Statistical Models

## But What Is OLS Really Doing?

> When errors are normally distributed, the OLS estimates of the regression coefficients are identical to the maximum likelihood estimates - minimizing squared residuals and maximizing the normal likelihood are the same problem.

To see why, we need to introduce **Maximum Likelihood (ML)**.

# 🎯 Estimation in Statistical Models

## Maximum Likelihood

A **probability density function** 🧐 describes the distribution of the data.

For a normal distribution: $$\mathrm{L}_i = \frac{1}{\sqrt{2\pi\sigma^2}} \mathrm{e}^{-.5\frac{(\mathrm{Y}_i-\mu)^2}{\sigma^2}}$$

$\mathrm{L}_i$ measures the relative probability of observing $\mathrm{Y}_i$ given a mean and variance.

# ML estimation

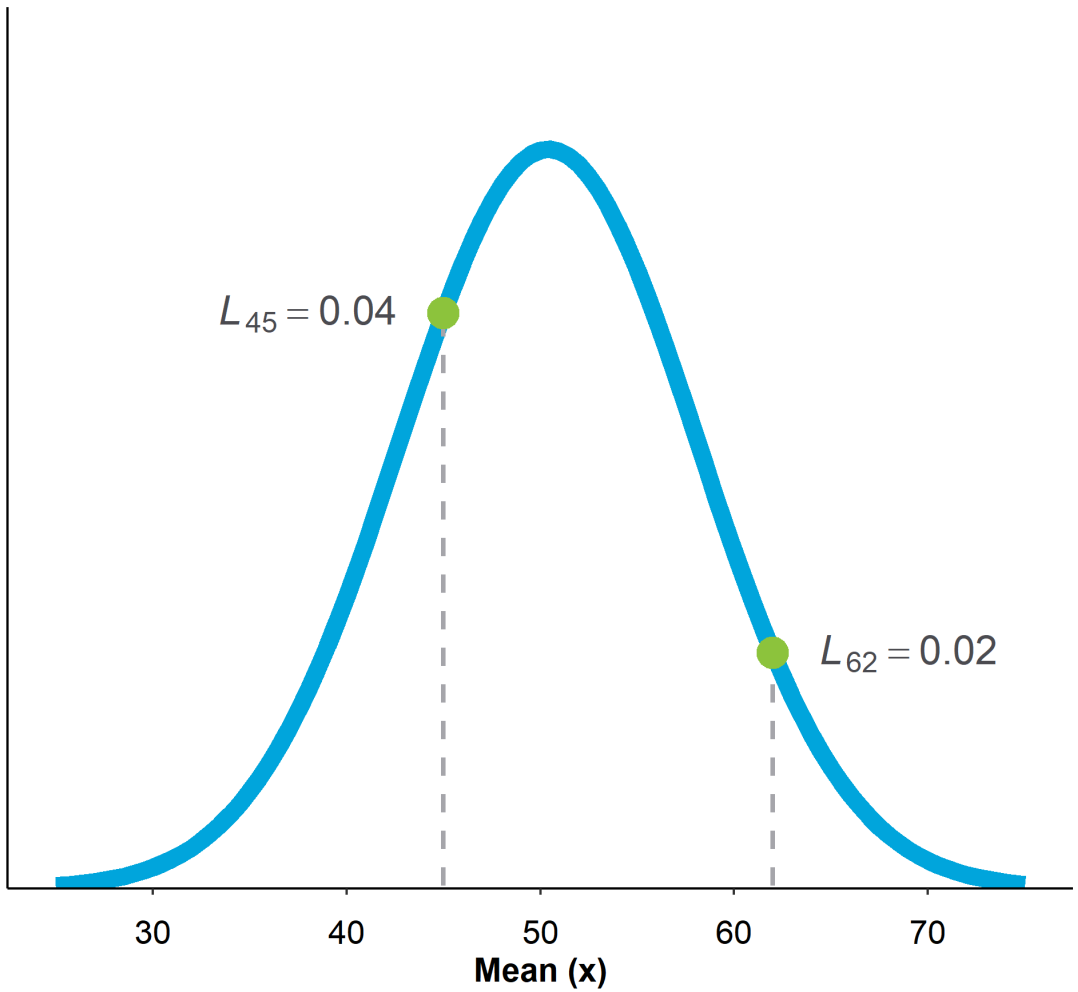
Applying the density function gives the relative probability $(L_i)$ of each score from this normal distribution.

## Complete data:

$$\mathrm{x}\ (\mu = 50.42, \sigma = 7.65)$$

**Largest $L_i$ is best!**

| ID | x | $L_i$ |
|---|---|---|
| 1 | 36.6 | 0.010201 |
| 2 | 41.8 | 0.027624 |
| 3 | 42.6 | 0.030908 |
| 4 | 43.1 | 0.032971 |
| 5 | 43.4 | 0.034205 |
| 6 | 44.2 | 0.037444 |
| 7 | 44.9 | 0.040166 |
| 8 | 46.3 | 0.045074 |
| 9 | 48.6 | 0.050658 |
| 10 | 49.0 | 0.051223 |
| 11 | 50.0 | 0.052038 |
| 12 | 51.6 | 0.051508 |
| 13 | 54.6 | 0.044915 |
| 14 | 54.8 | 0.044264 |
| 15 | 55.7 | 0.041102 |
| 16 | 57.2 | 0.035227 |
| 17 | 57.6 | 0.033589 |
| 18 | 60.3 | 0.022677 |
| 19 | 60.9 | 0.020433 |
| 20 | 65.3 | 0.007888 |


$L_{45} = 0.04$
$L_{62} = 0.02$
30
40
50
60
70
Mean (x)

# Maximum Likelihood

Multiple each $(L_i)$ to get sample likelihood.

0.0000000000000000000000000000000163666415977258

Fit of this data to $\mu$ = 50.42, $\sigma$ = 7.65

To avoid small numbers, we take the log of the likelihood.

Add each $logL_i$ to get sample loglikelihood.

-68.58

## Model Fit Indices use this!

$$\text{AIC} = -2 \times logL + 2k$$
$$\text{BIC} = -2 \times logL + k \times logN$$

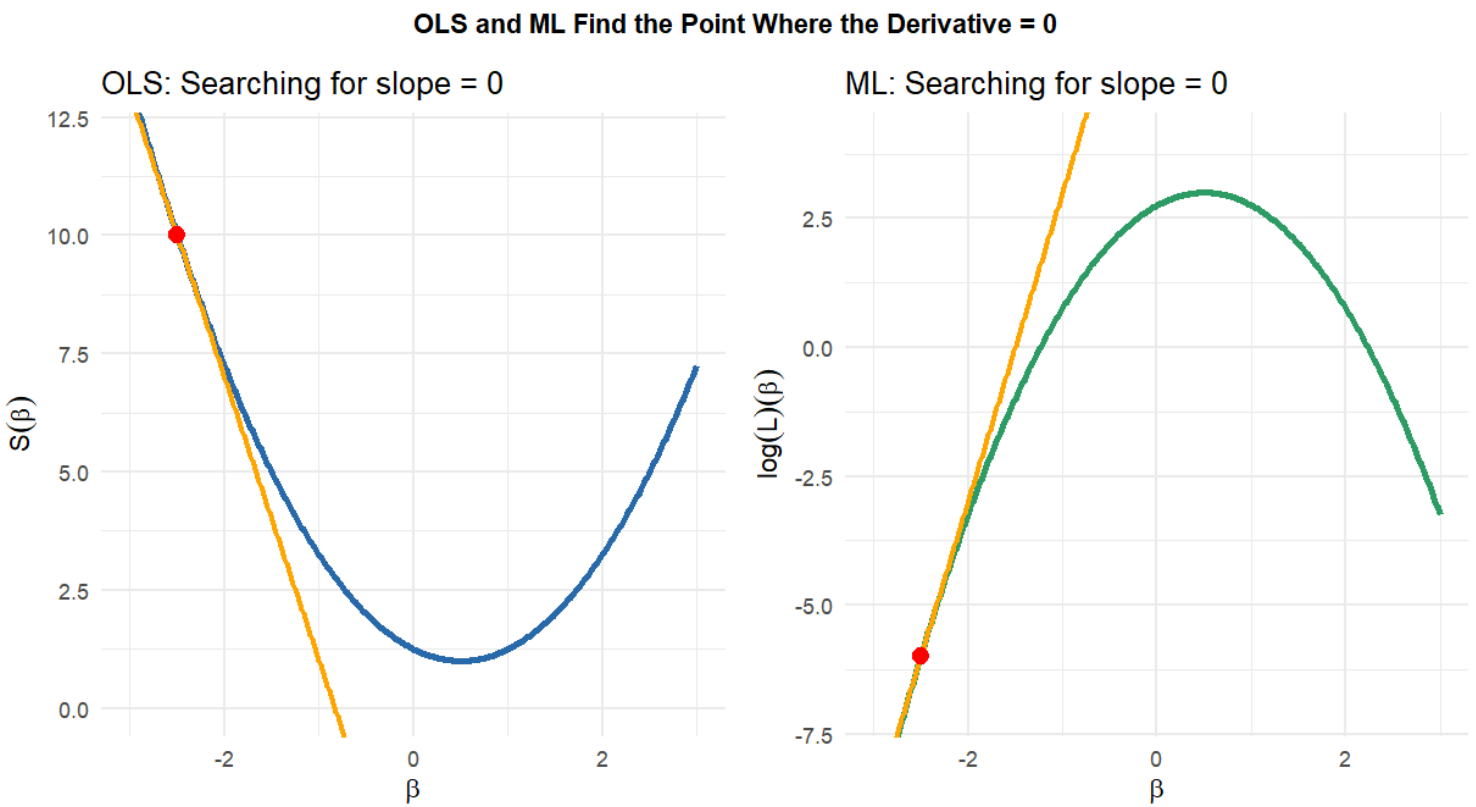
| ID | x | $L_i$ | $logL_i$ |
|---|---|---|---|
| 1 | 36.6 | 0.010201 | -4.585258 |
| 2 | 41.8 | 0.027624 | -3.589055 |
| 3 | 42.6 | 0.030908 | -3.476754 |
| 4 | 43.1 | 0.032971 | -3.412113 |
| 5 | 43.4 | 0.034205 | -3.375376 |
| 6 | 44.2 | 0.037444 | -3.284921 |
| 7 | 44.9 | 0.040166 | -3.214733 |
| 8 | 46.3 | 0.045074 | -3.099445 |
| 9 | 48.6 | 0.050658 | -2.982664 |
| 10 | 49.0 | 0.051223 | -2.97157 |
| 11 | 50.0 | 0.052038 | -2.955783 |
| 12 | 51.6 | 0.051508 | -2.966023 |
| 13 | 54.6 | 0.044915 | -3.102986 |
| 14 | 54.8 | 0.044264 | -3.117579 |
| 15 | 55.7 | 0.041102 | -3.191692 |
| 16 | 57.2 | 0.035227 | -3.345936 |
| 17 | 57.6 | 0.033589 | -3.393553 |
| 18 | 60.3 | 0.022677 | -3.786393 |
| 19 | 60.9 | 0.020433 | -3.890587 |
| 20 | 65.3 | 0.007888 | -4.842415 |

| ID | x | μ = 30 | μ = 40 | μ = 50 | μ = 60 | μ = 70 |
|---|---|---|---|---|---|---|
| 1 | 36.6 | -3.326 | -3.053 | -4.487 | -7.627 | -12.474 |
| 2 | 41.8 | -4.142 | -2.982 | -3.528 | -5.781 | -9.74 |
| 3 | 42.6 | -4.309 | -3.012 | -3.422 | -5.538 | -9.361 |
| 4 | 43.1 | -4.419 | -3.036 | -3.361 | -5.391 | -9.129 |
| 5 | 43.4 | -4.487 | -3.053 | -3.326 | -5.306 | -8.992 |
| 6 | 44.2 | -4.675 | -3.105 | -3.241 | -5.085 | -8.634 |
| 7 | 44.9 | -4.849 | -3.159 | -3.176 | -4.9 | -8.33 |
| 8 | 46.3 | -5.222 | -3.293 | -3.071 | -4.556 | -7.747 |
| 9 | 48.6 | -5.906 | -3.585 | -2.971 | -4.063 | -6.862 |
| 10 | 49 | -6.035 | -3.645 | -2.963 | -3.987 | -6.718 |
| 11 | 50 | -6.368 | -3.808 | -2.954 | -3.808 | -6.368 |
| 12 | 51.6 | -6.936 | -4.103 | -2.976 | -3.556 | -5.843 |
| 13 | 54.6 | -8.118 | -4.773 | -3.135 | -3.203 | -4.978 |
| 14 | 54.8 | -8.203 | -4.823 | -3.151 | -3.185 | -4.926 |
| 15 | 55.7 | -8.591 | -5.058 | -3.231 | -3.112 | -4.699 |
| 16 | 57.2 | -9.268 | -5.479 | -3.397 | -3.021 | -4.352 |
| 17 | 57.6 | -9.455 | -5.598 | -3.447 | -3.003 | -4.266 |
| 18 | 60.3 | -10.789 | -6.471 | -3.86 | -2.955 | -3.757 |
| 19 | 60.9 | -11.102 | -6.682 | -3.968 | -2.961 | -3.661 |
| 20 | 65.3 | -13.588 | -8.416 | -4.952 | -3.194 | -3.143 |
| | | -139.79 | -87.13 | -68.62 | -84.23 | -133.98 |

## ML estimation as an audition process

> We calculate the log-likelihood for every observation under each candidate μ, then sum them up. The total at the bottom of each column is that candidate's total log-likelihood score - its audition result.

- Here, the highest log-likelihood (least negative) wins!




> SSR is embedded inside the log-likelihood given errors are normal and independent with constant variance, so both methods locate the same optimal $\beta$.

# 🎯 Estimation in Statistical Models

## Why Move Beyond OLS?

OLS works only for:

- Normal outcomes
- One outcome at a time
- Independent observations
- Listwise deletion with missing data

# 🎯 Estimation in Statistical Models

## Maximum likelihood allows us to estimate:

- ✅ multilevel models
- ✅ generalized models (binary, count, etc.)
- ✅ models with missing data (FIML)
- ✅ multiple outcomes simultaneously

> ML is the general estimation framework for modern statistical models. We will use it!

# 🎯 Estimation in Statistical Models

## ML in Multilevel Models

Multilevel models estimate:

- Fixed effects
- Random effects
- Variance components

These parameters are estimated using **ML** or **REML**.

# 🎯 Estimation in Statistical Models

## ML vs. Restricted/Residual ML (REML)

| Method | Use Case | Pros / Notes |
|---|---|---|
| ML | Comparing models with different fixed effects | Random effects estimates may be slightly biased downward |
| REML | Comparing models with the same fixed effects, or interpreting random effects | Provides unbiased estimates of random effects var/covar<br><br>Cannot compare models with different fixed effects |

✅ Changing fixed effects? - Use ML to compare

✅ Changing random effects only? - Use REML to compare

✅ Final model for reporting? - Refit with REML for unbiased variance estimates.

# How to handle nested data?

# How to handle nested data?

Methods fall into three general categories based on how they handle nestedness.

- ⚠️ **Ignore:** disaggregation, aggregation
- 🙁 **Control:** ANCOVA, separate regressions, adjusting the results, fixed effects approach
- 🔍 **Model:** *Modeling* nestedness

> In fact, methods have shifted from *controlling clustering* (attempting to remove its influence) to **modeling it as something worth studying**! 🙌

# 🙈 🙉 🙊 What is disaggregation?

# 🙈 🙉 🙊 What is disaggregation?

> Disaggregation means **moving variables to a lower level** as if there is no clustering (see Hox, J., Moerbeek, M., & Van de Schoot, R., 2017).

- **The risk:** Inflated sample size and underestimated standard errors (false positives).

# 🙈 🙉 🙊 What is disaggregation?

- Level-1 units are the unit of analysis.
- Level-1 and level-2 predictors are included as if they were ordinary independent variables.

---

# 😟 Not recommended

> Violates the independence assumption which can lead to standard errors that are severely biased (usually underestimated).

- **Result:** Inflated Type I error rates and overconfident conclusions.

> Neglects between-cluster variation, effectively treating all

# 🙈 🙉 🙊 What is disaggregation?

| group | x | y |
|---|---|---|
| 1 | -9.1 | 6.7 |
| 1 | 20.5 | 9.7 |
| 1 | 36.7 | 8.9 |
| 2 | 13.1 | 6.4 |
| 2 | 24.8 | 8.3 |
| 2 | 21.7 | 5.0 |
| 3 | 47.9 | 4.3 |
| 3 | 45.5 | 6.6 |
| 3 | 34.9 | 4.6 |
| 4 | 45.1 | 2.1 |
| 4 | 84.1 | 5.3 |
| 4 | 26.3 | 1.0 |
| 5 | 80.9 | 0.1 |
| 5 | 61.5 | 0.8 |
| 5 | 77.4 | 5.5 |

## Example (Nested/Grouped Data)

- x = predictor, y = outcome
- 5 groups, showing first 3 rows
- 80 individuals per group (*N* = 400)
- Think: students in classrooms, patients in hospitals, repeated measures per person

$$\mathrm{y}_i = \beta_0 + \beta_1 \mathrm{x}_i + \mathrm{e}_i$$

# 🙈 🙉 🙊 What is disaggregation?

## R code:

```
pooled_lm <- lm(y ~ x, data = df)
summary(pooled_lm)
```

Table: Pooled OLS Regression with Centered Predictor

| term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|
| (Intercept) | 6.392 | 0.181 | 35.38 | 0.000 |
| x_c | 0.001 | 0.006 | 0.09 | 0.928 |

> In a linear regression predicting Y from X (mean-centered), the intercept was 6.392, representing the expected *Y* at the mean of *X*. The slope for *X* was 0.001, which was not statistically significant, $t(398) = 0.09$, $p = 0.928$. This analysis assumes that groups have no influence; the more group membership actually affects Y, the more misleading this result becomes.

# 🙈 🙊 🙉 What is disaggregation?

## Knowing the group matters!

- ⚠️ $e_i$ = individual + group variation

# 🙈 🙉 🙊 What is disaggregation?

| group | x | y | resid |
|---|---|---|---|
| 1 | -9.1 | 6.7 | 0.4 |
| 1 | 20.5 | 9.7 | 3.3 |
| 1 | 36.7 | 8.9 | 2.5 |
| 2 | 13.1 | 6.4 | 0 |
| 2 | 24.8 | 8.3 | 1.9 |
| 2 | 21.7 | 5.0 | -1.4 |
| 3 | 47.9 | 4.3 | -2.1 |
| 3 | 45.5 | 6.6 | 0.2 |
| 3 | 34.9 | 4.6 | -1.7 |
| 4 | 45.1 | 2.1 | -4.3 |
| 4 | 84.1 | 5.3 | -1.1 |
| 4 | 26.3 | 1.0 | -5.4 |
| 5 | 80.9 | 0.1 | -6.3 |
| 5 | 61.5 | 0.8 | -5.6 |
| 5 | 77.4 | 5.5 | -0.9 |

## Where the bias shows up:

The model overpredicts some groups and underpredicts others.

> Residual variance is inflated but the SE of the slope is too low because OLS treats all 400 points as independent. Accounting for strong within-group correlation shrinks the effective sample size from 400 to 12! 🤯

$$\mathrm{SE}(\hat{\beta}) = \sqrt{\frac{\sigma^2 \times \mathrm{VIF}}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \text{where} \mathrm{VIF} = 1 + (n_j - 1)\rho$$

Our estimated SE was 0.006, but adjusting for groups it should be 0.03.

# What is aggregation?

# What is aggregation?

- Level-2 units are the unit of analysis.
- Massive loss of information! 😜
- The results cannot be generalized to level-1 (*ecological fallacy*).

# What is aggregation?

R code:

```
agg_lm <- lm(y_mean ~ x_mean, data = group_means)
```

Table: Aggregated Regression (Group Means)

| term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|
| (Intercept) | 6.392 | 0.333 | 19.20 | 0.000 |
| x_mean | -0.100 | 0.015 | -6.43 | 0.008 |

> In a linear regression predicting group-mean Y from the group-mean X (both grand mean-centered). The slope for *X* was -0.1, which was statistically significant, $t(3) = -6.43$, $p = 0.008$, with a large effect size of $R^2 = 0.93$. This analysis is at the group level, and cannot be assumed to apply to individuals within those groups.

# 🛠 The fixed effects approach?

# 🛠 The fixed effects approach?

## Include group as a fixed effect via dummy coding.

- Separate intercept for each group
- Slopes equal (unless interactions added)
- Many parameters (50 groups = 49 dummy codes) 😟

> Imagine 5 specific hospitals (Seattle Children's, CHOP, and so on) and we chose these hospitals deliberately because we want to study *these* hospitals specifically. Our estimates tell us how much each hospital's intercept differs from the reference hospital, and we never intend to say anything beyond these 5 hospitals anyway. Because slopes are fixed (parallel lines), the effect of *X* represents the common within-hospital relationship across all hospitals.

# 🛠 The fixed effects approach?

### R code:

```
fixed_lm <- lm(y ~ x + factor(group), data = df)
```

Table: Fixed-effects Regression (dummy codes)

| term | estimate | std.error | statistic | p.value |
|---|---|---|---|---|
| (Intercept) | 8.1 | 0.211 | 38.4 | 0 |
| x | 0.1 | 0.004 | 23.6 | 0 |
| factor(group)2 | -2.9 | 0.307 | -9.4 | 0 |
| factor(group)3 | -6.4 | 0.327 | -19.5 | 0 |
| factor(group)4 | -9.4 | 0.363 | -25.8 | 0 |
| factor(group)5 | -12.2 | 0.395 | -30.9 | 0 |

This approach absorbs between-group variation and focuses on within-group associations, producing more realistic SEs than naive pooled OLS.

# 🛠 The fixed effects approach?

## 1️⃣ No bias from omitted group effects

- The over/under-prediction problem from pooled OLS goes away because each group gets its own intercept.

## 2️⃣ No design-effect inflation from clustering

- The ICC is controlled by the group terms

## 3️⃣ Effective sample size is 396 after adjustment.

*Assumes fixed slopes unless you add interactions.*

# 📊 ANCOVA approach?

# 📊 ANCOVA approach?

## Same model - different interpretation and motivation.

- Cannot model varying slopes across groups
- Cannot include level-2 predictors of group effects
- Group differences are controlled, not modeled

# Post-adjustment?

# 🧩 Post-adjustment?

## If nesting biases standard errors, adjust them directly.

### R code:

```
pooled_lm <- lm(y ~ x_c, data = df)
cluster_se <- sandwich::vcovCL(pooled_lm, cluster = df$group, type = "HC1")
```

Table: Fixed-effects Regression with Cluster-Robust SEs

| term | estimate | std.error | t_value | p_value |
|---|---|---|---|---|
| (Intercept) | 6.392 | 1.115 | 5.73 | 0.000 |
| x_c | 0.001 | 0.024 | 0.02 | 0.982 |

- The new SE = 0.02 up from 0.006
- Does not model cluster-specific effects
- Generally not recommended for multilevel data

# Separate regressions?

# Separate regressions?

## Analyze each level-2 unit separately.

- 5 groups = 5 models

# 🔀 Seperate regressions?

## 5 separate regressions ➡️ get the variance across groups

Table: Separate Regressions per Group: Intercepts and Slopes

| group | intercept | slope |
|---|---|---|
| 1 | 7.98 | 0.11 |
| 2 | 5.76 | 0.09 |
| 3 | 1.27 | 0.11 |
| 4 | -1.70 | 0.11 |
| 5 | -3.27 | 0.09 |

Table: Variance and Covariance Across Groups (equal weight)

| I_Var | S_Var | IS_Cov |
|---|---|---|
| 22.995 | 0.00017 | 0.007 |

# 🔀 Separate regressions?

- Examine variance of intercepts & slopes
- All clusters are weighted equally
- Does **not partition variance**
- Involves estimating a **large number of parameters!**

> 50 hospitals = 50 separate regressions = 500 parameters - this requires a ton of data! 😔

# 🤨 Which Effect Is Correct?

**What happens with the same data can be confusing.**

> Ignoring clustering changes the estimated relationship.

**We need a model that separates within- and between-group effects.**

# The Solution: Multilevel Models

# The Solution: Multilevel Models

**Multilevel models do something different:**

they *simultaneously* model variation **within** and **between** clusters.

# The Solution: Multilevel Models

**Model the structure - don't ignore it, don't destroy it.**

*All the approaches so far fall short in some way:*

- **Disaggregation:** Easy but biased; treats all observations as exchangeable
- **Aggregation:** Only higher level data; different questions
- **Dummy codes:** Assumes *one* shared story across clusters, unwieldy
- **Post-adjustment:** Corrects bias but ignores cluster variation
- **Separate regressions:** Parameter-hungry and can't generalize

# The Solution: Multilevel Models

## Model dependency across observations

- Longitudinal, clustered, or cross-classified data? ✅ No problem!
- Tailor your model to the sources of correlation in your data

## Include categorical or continuous predictors at any level

- Time-varying, person-level, or group-level predictors
- Explore why observations are correlated, don't just control for it

## Handle unbalanced or missing data

- Not every person needs the same number of measurements
- Missing data? ✅ No problem!

# Questions? 🤔

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Next: *Diving into Multilevel Models (MLM)*